

Topic 5

Redox Reactions, Chemical Cells and Electrolysis

Chemistry in Context



Hand warmer Whether you are camping, hiking or skiing, you should always carry a hand warmer with you. A hand warmer not only keeps your hands warm for outdoor recreational activities, it can also be a life saver to keep your hands or feet warm during an emergency when you may be stuck in the cold for an extended period of time. One type of hand warmer is made up of a mixture of iron, activated carbon, water and other salts wrapped in a pack that keeps out oxygen. Unwrapping the warmer allows oxygen to oxidize the iron and heat is released from the reaction. This reaction is similar to the rusting of iron.

Do you recall?

1 What is the difference between rusting and corrosion?

2 What is the chemical formula of rust?

3 What is the name of a summary table that shows the order of reactivity of metals? Can you recall the order?

Refer to Teacher's Resources Book page 32.

In this topic, you are going to find out more about oxidation.

Unit

18 Chemical cells in daily life

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Different types of chemical cells
- Choosing a chemical cell for a particular use
- Environmental impact of using chemical cells

18.1 Electricity from chemical reactions

When we add some magnesium powder to a beaker of copper(II) sulphate solution, the temperature of the mixture rises. In this reaction, the chemical energy of the reactants is converted into heat.

We can also convert the chemical energy of the reactants into electrical energy.

✓ A **chemical cell** is a device in which chemical energy is converted into electrical energy.

$$\text{chemical energy} \xrightarrow{\text{chemical cell}} \text{electrical energy}$$

A chemical cell consists of two different metals and an **electrolyte**. The electrolyte contains mobile ions and thus can conduct electricity.

Fig. 18.1 shows a chemical cell. It is made up of magnesium and copper. Copper(II) sulphate solution is the electrolyte.

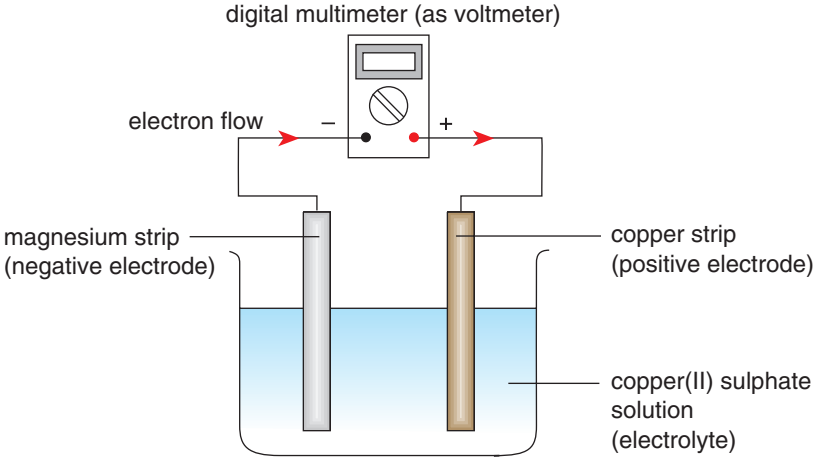


Fig. 18.1 A magnesium-copper chemical cell

chemical cell 化學電池　　electrolyte 電解質

In the magnesium-copper chemical cell, electrons flow from the magnesium strip to the copper strip in the external circuit. This flow of electrons forms an electric current. The magnesium strip is the **negative electrode**. The copper strip is the **positive electrode**. Ions can move freely in the copper(II) sulphate solution.

We will discuss the reason why electrons flow from magnesium to copper in Unit 19.

By convention, current flows from the positive electrode in the cell to the negative electrode. Actually, it is the electrons that are flowing. The flow of current and that of electrons are opposite in direction.

We can measure the voltage by a *voltmeter* or a *digital multimeter* (Figs. 18.2–18.3). When the positive electrode of the cell is connected to the positive terminal of the voltmeter and the negative electrode of the cell is connected to the negative terminal of the voltmeter, the voltmeter gives a positive voltage.

A digital multimeter is preferred as it gives more accurate readings.



Fig. 18.2 A voltmeter



Fig. 18.3 A digital multimeter

18.2 Different types of chemical cells

There are two main classes of chemical cells.

CE 2011 Paper 2 Q32
DSE 2014 Paper 1A Q16

Primary cells are not rechargeable. They provide electricity until the reactants have reacted to such an extent that the voltage drops to a certain level. The cell is then 'flat' and discarded. Zinc-carbon cell, alkaline manganese cell and silver oxide cell are primary cells.

A cell can be regarded as 'flat' (run-out) when its voltage falls to 0.8 V.

Secondary cells are rechargeable. The cell reaction can be reversed during recharging — the reactants in the cell are regenerated and the cell can be used again. Lithium ion cell, nickel metal hydride (NiMH) cell and lead-acid accumulator are secondary cells.

DSE 2013 Paper 1A Q21

negative electrode 負電極　positive electrode 正電極　voltmeter 伏特計　digital multimeter 數字萬用電錶
primary cell 原電池　secondary cell 二級電池

Fig. 18.4 shows different types of chemical cells commonly used.



Fig. 18.4 Different types of chemical cells



Fig. 18.5 A 9 V battery has six cells, each with a voltage of 1.5 V

18.3 Terms related to chemical cells

The following terms are related to chemical cells:

- A **battery** consists of two or more connected cells. Fig. 18.5 shows that a 9 V battery which consists of six 1.5 V cells connected in series.
- The **negative electrode** is the electrode from which electrons flow into the external circuit.
- The **positive electrode** is the electrode into which electrons flow from the external circuit.
- An **electrolyte** is the medium which allows ions to flow between the two electrodes of a cell.
- **Cell capacity** is the quantity of electricity a cell can deliver under specified conditions. It is usually expressed in milliampere-hours (mAh).
- **Discharge** is the conversion of the chemical energy of a cell into electrical energy.
- **Service life** is the length of time a cell can remain in use, before its voltage drops to a certain value (usually 0.8 V).
- **Cycle life** is the number of times that a secondary cell can be charged and discharged before it can no longer deliver a sufficient amount of energy.
- **Shelf life** is the length of time a cell can remain in storage (at 21 °C) without losing more than 10% of its capacity.

The negative electrode of a chemical cell may also be called the anode.

The positive electrode of a chemical cell may also be called the cathode.

Shelf life is more of a concern for primary cells than secondary cells as the latter can be recharged to raise its capacity.

battery 電池組　cell capacity 電池容量　discharge 放電　service life 有效壽命　cycle life 循環壽命　shelf life 存放期

There are two common cell shapes: cylindrical and button. Fig. 18.6 shows some common cylindrical cells of different sizes.



Fig. 18.6 Some common cylindrical cells of different sizes

18.4 Zinc-carbon cell

Zinc-carbon cells (Fig. 18.7) are the most commonly used chemical cells. They are often used in torches, small radios and remote control devices.



Fig. 18.7 Zinc-carbon cells

Fig. 18.8 shows a simplified cross-section of a zinc-carbon cell. A zinc-carbon cell contains a carbon rod surrounded by a mixture of powdered carbon and manganese(IV) oxide. The carbon rod is the positive electrode. The cylindrical casing is made of zinc, which serves as the negative electrode. The electrolyte is a moist paste of ammonium chloride. The maximum voltage of a zinc-carbon cell is 1.5 V, regardless of its size. N1

CE 2006 Paper 2 Q32 CE 2007 Paper 2 Q44 CE 2011 Paper 2 Q32

We will discuss the chemistry of a zinc-carbon cell in Unit 21.

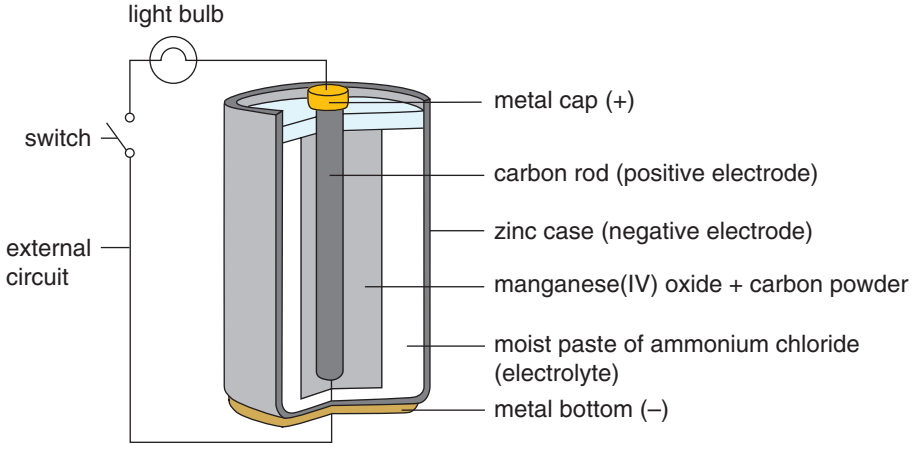


Fig. 18.8 Simplified cross-section of a zinc-carbon cell

N1 Examination questions may ask about the materials used for making the anode and cathode of a zinc-carbon cell.

In this case, the anode is made of zinc while the cathode is made of carbon / graphite.

zinc-carbon cells 鋅碳電池

The circuit is complete only when we close the switch in the external circuit. Electrons then flow from the zinc electrode to the carbon electrode through the external wiring, the closed switch and the bulb.

Reference Website

Refer to the following website for an animation illustrating how chemical cells power a flashlight:
http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/animations/flashlightV8.html
(accessed July 2014)

18.5 Alkaline manganese cell



Fig. 18.9 Alkaline manganese cells

Alkaline manganese cells (Fig. 18.9) can deliver higher currents than zinc-carbon cells can do. They can also deliver a greater quantity of electricity than zinc-carbon cells of the same size can do. Hence alkaline manganese cells are preferred for appliances requiring high currents (e.g. photographic electronic flash units) or high total cell capacity (e.g. motorized toys).

Compared with zinc-carbon cells, alkaline manganese cells have several advantages:

- have higher energy density[◀];
- show good performance at all drain rates;
- give steadier voltage over discharge;
- show good low-temperature performance;
- have longer shelf life.

◀ The energy density is a measure of the amount of energy per unit weight or per unit volume which can be stored in a cell.

The negative electrode of an alkaline manganese cell is zinc powder. The positive electrode is manganese(IV) oxide. The electrolyte is potassium hydroxide[◀], which soaks both the zinc powder and the manganese(IV) oxide. The maximum voltage of this cell is 1.5 V. N2

◀ Do NOT attempt to open an alkaline manganese cell because potassium hydroxide is corrosive.

N2 At the negative electrode (oxidation): $Zn(s) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(l) + 2e^-$
At the positive electrode (reduction): $2MnO_2(s) + H_2O(l) + 2e^- \longrightarrow Mn_2O_3(s) + 2OH^-(aq)$
The overall cell reaction is: $Zn(s) + 2MnO_2(s) \longrightarrow ZnO(s) + Mn_2O_3(s)$

18.6 Silver oxide cell

CE 2008 Paper 2 Q23

In recent years, primary cells have been developed with improved voltage and reduced size. *'Button' cells* (Fig. 18.10) are used in heart *pacemakers*, hearing aids, watches and calculators. They have the following advantages:



Fig. 18.10 Silver oxide button cells

- lightweight and small;
- give a constant voltage over discharge.

alkaline manganese cell 鹼性錳電池 button cell 鈕型電池 pacemaker 起搏器

Silver oxide cell is a type of button cell. The negative electrode is zinc powder. The positive electrode is silver oxide. These materials are separated by an electrolyte containing potassium hydroxide. The maximum voltage of this cell is 1.5 V. N3

Table 18.1 summarizes the characteristics of the three types of primary cells discussed above.

N3 At the negative electrode (oxidation): $Zn(s) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(l) + 2e^-$
At the positive electrode (reduction): $Ag_2O(s) + H_2O(l) + 2e^- \longrightarrow 2Ag(s) + 2OH^-(aq)$
The overall cell reaction is: $Zn(s) + Ag_2O(s) \longrightarrow ZnO(s) + 2Ag(s)$

Table 18.1

Characteristics of the three types of primary cells

	Zinc-carbon cell	Alkaline manganese cell	Silver oxide cell
Material of negative electrode	zinc	zinc	zinc
Material of positive electrode	carbon	manganese(IV) oxide	silver oxide
Material of electrolyte	ammonium chloride	potassium hydroxide	potassium hydroxide
Maximum voltage (V)	1.5	1.5	1.5 N4
Advantages	• low cost • long shelf life	• good performance at all drain rates • a steadier voltage over discharge • good low-temperature performance • long shelf life	• lightweight • small size • a constant voltage over discharge
Limitation(s)	• poor performance in high-drained devices • output voltage decreases as the cell is discharged • poor low-temperature performance	• more expensive than zinc-carbon cell	• the most expensive among the primary cells

Reference Website

Refer to the following websites for application manuals and guidebooks of different types of cells:
– http://data.energizer.com/Static.aspx?Name=AppManuals (accessed July 2014)
– http://support.radioshack.com/support_tutorials/batteries/batgd-b.htm (accessed July 2014)

N4 The size of a chemical cell does not affect its voltage.

silver oxide cell 氧化銀電池

18.7 Lithium ion cell

Lithium ion cells (Figs 18.11–18.12) offer the greatest amount of energy for a given size or weight of all commercial rechargeable cells. Lithium ion cells are commonly used in mobile phones and other portable electronic devices.

The negative electrode of a lithium ion cell consists of lithium atoms lying between graphite sheets. The positive electrode is made of a lithium metal oxide, such as lithium cobalt oxide or lithium manganese oxide. Since lithium reacts violently with water, the electrolyte is composed of a lithium salt dissolved in an organic solvent. The maximum voltage of this cell is 3.7 V. N5



Fig. 18.11 Lithium ion cells



Fig. 18.12 Battery chargers for some lithium ion cells

Lithium ion cells are so popular because they have a number of advantages:

- have very high energy density;
- are much lighter than other types of secondary cells of the same size;
- have low self-discharge rate◀.

Safety concerns of lithium ion cells

Lithium ion cells cannot take high charge and discharge currents. Trying to force a rapid charge or loading the cell with excess discharge current may overheat the pack and lead to an explosion.

Do you know

Different types of lithium ion cells

Most lithium ion cells used for portable applications are cobalt-based. The cobalt lithium ion cell was first introduced in 1991. In spite of its great popularity, the cell has some limitations. It cannot take high charge and discharge currents. Another drawback is the increase of the internal resistance that occurs with cycling and aging.

In 1996, scientists succeeded in using lithium manganese oxide as the positive electrode material. The internal resistance of this cell stays low with cycling and aging. This characteristic allows for fast-charging and high-current discharging. Besides, in comparison with cobalt, manganese is a safer and more environmentally benign electrode material. These cells are often used in power tools and medical equipment.

◀ Most cells undergo slow self-discharge when left standing on open-circuit.

N5 Refer to Teacher's Resources Book page 18.

lithium ion cell 鋰離子電池

18.8 Nickel metal hydride (NiMH) cell



Fig. 18.13 Nickel metal hydride (NiMH) cells and charger

A **nickel metal hydride** (NiMH) **cell** (Fig. 18.13) consists of a hydrogen absorbing alloy negative electrode, a nickel(II) hydroxide positive electrode and a potassium hydroxide electrolyte. The maximum cell voltage is 1.2 V. N6

Nickel metal hydride cells have several advantages:

- have high energy density;
- show good performance in high-drained devices;
- have a wide operating temperature range.

However, these cells have a high self-discharge rate.

18.9 Lead-acid accumulator



Fig. 18.14 A lead-acid accumulator

A **lead-acid accumulator** (Fig. 18.14) is called an accumulator as it can accumulate or store electrical energy. It is commonly used in cars. A 12 V accumulator normally consists of six cells joined in series. The cells are contained in a heavy duty external casing.

The negative electrode of each cell is made of lead plates and the positive electrode is made of titanium plates coated with lead(IV) oxide. The electrolyte is sulphuric acid. The voltage of each cell is 2 V. N7

Lead-acid accumulators were invented in 1859. They remain the choice for automotive SLI (Starting, Lighting and Ignition) applications.

Lead-acid accumulators have several advantages:

- robust;
- tolerant to overcharging;
- capable of delivering high currents;
- low cost.

N7 At the negative electrode (oxidation):

$$Pb(s) + SO_4^{2-}(aq) \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} PbSO_4(s) + 2e^-$$

At the positive electrode (reduction):

$$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} PbSO_4(s) + 2H_2O(l)$$

N6 At the negative electrode (oxidation):

$$MH(s) + OH^-(aq) \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} M(s) + H_2O(l) + e^-$$

where MH represents a hydrogen absorbing alloy.

At the positive electrode (reduction):

$$NiO(OH)(s) + H_2O(l) + e^- \underset{\text{charging}}{\overset{\text{discharging}}{\rightleftharpoons}} Ni(OH)_2(s) + OH^-(aq)$$

nickel metal hydride cell 鎳金屬氫化物電池　　lead-acid accumulator 鉛酸蓄電池

However, lead-acid accumulators are heavy and bulky. Lead and lead compounds are also toxic.

Table 18.2 summarizes the characteristics of the three types of secondary cell discussed above.

Table 18.2

Characteristics of three types of secondary cell

	Lithium ion cell	NiMH cell	Lead-acid accumulator
Materials of negative electrode	lithium atoms lying between graphite sheets	hydrogen-absorbing alloys	lead plates
Material of positive electrode	a lithium metal oxide (such as lithium cobalt oxide or lithium manganese oxide)	nickel(II) hydroxide	titanium plates coated with lead(IV) oxide
Material of electrolyte	a lithium salt dissolved in an organic solvent	potassium hydroxide	sulphuric acid
Maximum voltage (V)	3.7	1.2	2
Weight	light	moderate	very heavy
Energy density	very high (about 4 times of a lead-acid accumulator)	high (about 2 times of a lead-acid accumulator)	lowest among the three
Usage	electric razors, electric toothbrushes, medical equipment, mobile phones, laptop computers	electric razors, electric toothbrushes, video cameras, emergency backup lighting, medical equipment, electric vehicles	for automotive SLI (Starting, Lighting and Ignition) applications, wheelchairs, golf carts
Advantages	• high energy density • low self-discharge rate	• good performance in high-drained devices • wide operating temperature range	• robust • low cost
Limitations	• very high initial cost • relatively low discharge current • safety concerns	• high initial cost • high rate self-discharge	• heavy and bulky • lead and lead compounds are toxic

18.10 Choosing a chemical cell for a particular use

When choosing a chemical cell for a particular use, we need to consider the following aspects:

- price;
- voltage;
- capacity of the cell;
- whether the cell is rechargeable;
- whether the cell is able to supply a steady current;
- whether a large or small current is required;
- whether the cell is used continuously or *intermittently*;
- whether the cell is required to operate at low temperatures;
- shelf life; and
- risk of leakage.

When choosing a chemical cell, an important decision to make is whether the application merits the purchase of a primary or a secondary cell.

Which cell is the most suitable for mobile phones?



Fig. 18.15 Lithium ion cells are used to power modern mobile phones

Early mobile phones were powered with nickel-based cells but modern phones are equipped with lithium ion cells (Fig. 18.15). When compared with other types of cell, lithium ion cell weighs less, offers a higher energy density, lasts long enough to span the typical life of the mobile phone and contains no toxic metals.

A lithium ion cell has a high cell voltage (3.7 V). Hence one cell would be sufficient to make up a battery pack. A battery pack using 1.2 V nickel-based cells would require three cells connected in series.

intermittent 間歇的

Which cell is the most suitable for automotive use?

Although nickel metal hydride cells have been almost completely replaced by lithium ion cells in mobile electronic devices, they are still used in automotives. Their wide operating temperature range makes them ideal for such applications. Nickel metal hydride cells are also safer than lithium-based cells. This property is very important for automotive applications.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Decision Making

The following table shows the information about three different types of AA-sized cylindrical cell.

Type	Price per cell (HKD)	Approximate service life			
		Torch (continuous lighting)	Camera flash unit	Portable CD player	Radio
Zinc-carbon	2.00	0.5 hours	15 times	3 hours	50 hours
Alkaline manganese	6.00	1.7 hours	200 times	12 hours	140 hours
Nickel metal hydride	24.0	service life is not given here as this is unimportant for a rechargeable cell			

Cost of a battery charger = HKD 115

The cost for charging an AA-sized nickel metal hydride cell is very low, usually about HKD 0.2–0.3.

Based on the given information, decide which type of cell you would use in each of the following devices. Explain your choice in each case.

a) Torch

b) Camera flash unit

c) Portable CD player

d) Radio

Refer to Teacher's Resources Book page 32.

18.11 Environmental impact of using chemical cells

Disposal of 'flat' cells in fire may lead to explosion.

Chemical cells are commonly used nowadays. Many cells are useless when they become 'flat'. Disposal of these 'flat' cells presents an environmental problem.

Heavy metal components of chemical cells can be toxic to plants, animals and humans when discarded. Cadmium, lead and mercury are the heavy metal components that cause the most environmental concerns.

At one time, small amounts of mercury were used as an additive in zinc-carbon and alkaline cells. In response to environmental concerns, manufacturers have developed new technologies that eliminate the use of mercury (Fig. 18.16).



Fig. 18.16 Alkaline chemical cells without mercury

Using secondary cells is more cost efficient and environmentally friendly than using primary cells because secondary cells can be recharged and reused. Using secondary cells reduces the quantity of discarded cells in landfills.

Have you mastered?

Key terms

chemical cell 化學電池 2 electrolyte 電解質 2 negative electrode 負電極 3

positive electrode 正電極 3 voltmeter 伏特計 3 digital multimeter 數字萬用電錶 3

primary cells 原電池 3 secondary cells 二級電池 3 battery 電池組 4

cell capacity 電池容量 4 discharge 放電 4 service life 有效壽命 4 cycle life 循環壽命 4

shelf life 存放期 4 zinc-carbon cell 鋅碳電池 5 alkaline manganese cell 鹼性錳電池 6

silver oxide cell 氧化銀電池 7 lithium ion cell 鋰離子電池 8

nickel metal hydride cell 鎳金屬氫化物電池 9 lead-acid accumulator 鉛酸蓄電池 9

Checklist

After studying this unit, you should be able to

- ☐ distinguish between primary and secondary cells;
- ☐ describe the characteristics of common primary and secondary cells (zinc-carbon cell, alkaline manganese cell, silver oxide cell, lithium ion cell, nickel metal hydride cell and lead-acid accumulator);
- ☐ justify the uses of different chemical cells for particular purposes;
- ☐ understand the environmental impact of using chemical cells.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 A chemical cell is a device in which chemical energy is converted into electrical energy. It consists of two different metals and an electrolyte.

2 There are two main classes of chemical cells: primary cells and secondary cells. Primary cells are not rechargeable. Secondary cells are rechargeable.

3 The following table summarizes the characteristics of six common chemical cells.

	Zinc-carbon	Alkaline manganese	Silver oxide	Lithium ion	Nickel metal hydride	Lead-acid accumulator
Type	primary	primary	primary	secondary	secondary	secondary
Material of negative electrode	zinc	zinc	zinc	lithium atoms lying between graphite sheets	hydrogen-absorbing alloy	lead
Material of positive electrode	carbon	manganese(IV) oxide	silver oxide	lithium metal oxide	nickel(II) hydroxide	titanium plates coated with lead(IV) oxide
Material of electrolyte	ammonium chloride	potassium hydroxide	potassium hydroxide	lithium salt dissolved in an organic solvent	potassium hydroxide	sulphuric acid
Maximum voltage (V)	1.5	1.5	1.5	3.7	1.2	2
Advantage(s)	• low cost • long shelf life	• good performance at all drain rates • steadier voltage over discharge • good low-temperature performance • long shelf life	• lightweight • small size • a constant voltage over discharge	• high energy density • low self-discharge rate	• good performance in high-drained devices • wide operating temperature range	• robust • low cost
Limitation(s)	• poor performance in high-drained devices • output voltage decreases as the cell is discharged • poor low-temperature performance	• more expensive than zinc-carbon cell	• the most expensive among primary cells	• very high initial cost • relatively low discharge current • safety concerns	• high initial cost • relatively high of self-discharge rate	• heavy and bulky • lead and lead compounds are toxic

4 Disposal of chemical cells may cause environmental problems. Heavy metal components of chemical cells can be toxic to plants, animals and humans.

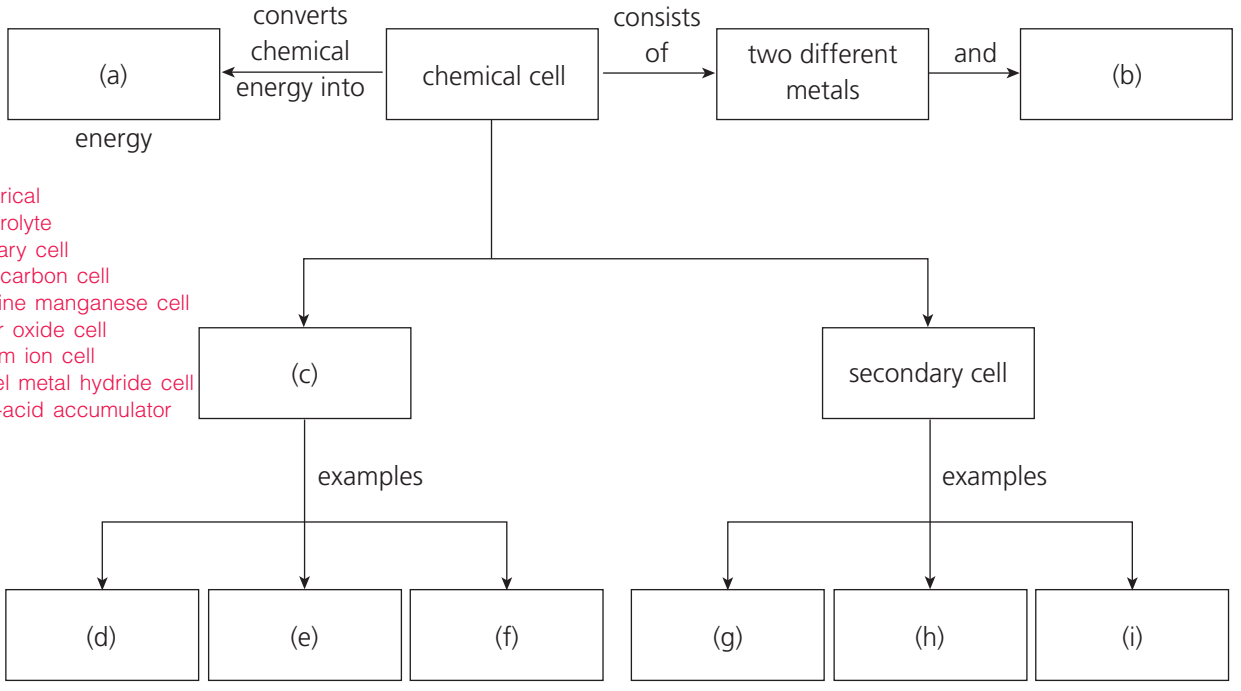
Unit Exercise

Refer to Teacher's Resources Book pages 32–35.

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.



a) electrical
b) electrolyte
c) primary cell
d) zinc-carbon cell
e) alkaline manganese cell
f) silver oxide cell
g) lithium ion cell
h) nickel metal hydride cell
i) lead-acid accumulator

2 a) The voltage of a chemical cell can be measured by a voltmeter or a digital multimeter .

b) A battery consists of two or more connected cells.

c) In a chemical cell, the negative electrode is the electrode from which electrons flow into the external circuit. The positive electrode is the electrode into which electrons flow from the external circuit.

d) A / an electrolyte is the medium which allows ions to flow between the two electrodes of a chemical cell.

e) The period of useful life of a chemical cell before its voltage drops to a certain value is its service life.

f) The duration of storage at the end of which a chemical cell retains 90% capacity is its shelf life.

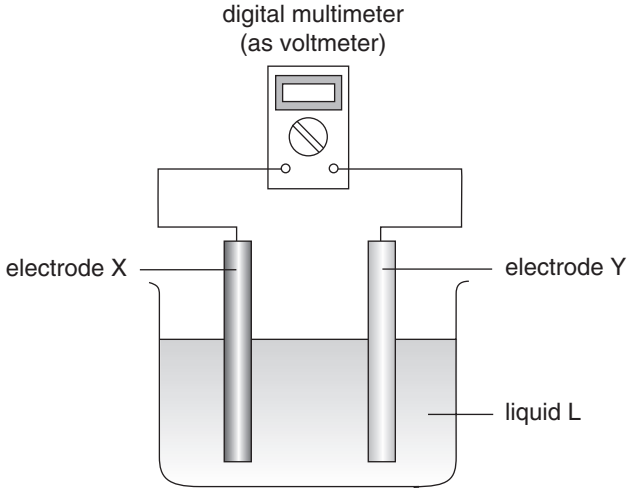
3 Complete the following table for the different types of chemical cells.

Chemical cell	Rechargeable or not?	Material of negative electrode	Material of positive electrode	Material of electrolyte	Maximum voltage (V)
Zinc-carbon cell	no	zinc	carbon	ammonium chloride	1.5
Alkaline manganese cell	no	zinc	manganese(IV) oxide	potassium hydroxide	1.5
Silver oxide cell	no	zinc	silver oxide	potassium hydroxide	1.5
Lithium ion cell	yes	lithium atoms lying between graphite sheets	lithium metal oxide	lithium salt dissolved in an organic solvent	3.7
Nickel metal hydride cell	yes	hydrogen-absorbing alloy	nickel(II) hydroxide	potassium hydroxide	1.2
Lead-acid accumulator	yes	lead	titanium plates coated with lead(IV) oxide	sulphuric acid	2

Part II Multiple choice questions

Refer to Teacher's Resources Book for explanations.

4 Consider the set-up shown below:



In which of the following cases will there be electrons flowing in the external circuit?

	Electrode X	Electrode Y	Liquid L
(A)	copper	zinc	dilute hydrochloric acid
B	copper	copper	dilute hydrochloric acid
C	silver	copper	distillled water
D	silver	zinc	ethanol

5 Which of the following statements concerning zinc-carbon cell is correct?

A Zinc-carbon cell is rechargeable.
(B) The positive electrode of zinc-carbon cell is a carbon rod.
C A zinc-carbon cell is a dry cell which does not contain water.
D A zinc-carbon cell of larger size produces a higher voltage than a smaller one.

(HKCEE, Paper 2, 2011, 32)

6 Which of the following statements about a lithium ion cell is correct?

A The maximum voltage of a lithium ion cell is 2 V.
B The positive electrode of a lithium ion cell consists of lithium atoms lying between graphite sheets.
(C) A lithium ion cell is much lighter than other types of secondary cells of the same size.
D A lithium ion cell can take high charging currents.

7 Which of the following combinations is correct for a nickel metal hydride secondary cell?

	Negative electrode	Positive electrode	Electrolyte
(A)	hydrogen absorbing alloys	nickel(II) hydroxide	potassium hydroxide
B	hydrogen absorbing alloys	potassium hydroxide	nickel(II) hydroxide
C	potassium hydroxide	hydrogen absorbing alloys	nickel(II) hydroxide
D	nickel(II) hydroxide	hydrogen absorbing alloys	potassium hydroxide

8 Which of the following statements about nickel metal hydride cells is INCORRECT?

A They have a high energy density.
B They are rechargeable.
(C) They have a low self-discharge rate.
D They have a wide operating temperature range.

9 Which of the following statements about a lead-acid accumulator is correct?

A The voltage of each cell in the accumulator is 12 V.
B It uses hydrochloric acid as an electrolyte.
C It has a very high energy density.
(D) It is capable of delivering high currents.

10 Which of the following is / are secondary cell(s)?

(1) Alkaline manganese cell
(2) Lithium ion cell
(3) Nickel metal hydride cell

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKDSE, Paper 1A, 2013, 21)

11 Which of the following statements about a zinc-carbon cell is / are correct?

(1) Manganese(IV) oxide acts as an electrolyte.
(2) The negative electrode is the zinc case.
(3) They show good performance at all drain rates.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

12 Which of the following are advantage(s) of using alkaline manganese cells over zinc-carbon cells in motorized toys?

(1) Alkaline manganese cells have longer life time.
(2) Alkaline manganese cells have a higher maximum voltage.
(3) Alkaline manganese cells give a steadier voltage over discharge.

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

13 Which of the following is / are related to the use of silver oxide cells in watches?

(1) Silver oxide cells are rechargeable.
(2) Silver oxide cells are small in size.
(3) Silver is an expensive metal.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2008, 23)

14 Which of the following statements about lithium ion batteries is / are correct?

(1) In lithium ion batteries, the electrolyte is a lithium salt in water.
(2) Lithium ion batteries are rechargeable.
(3) The disposal of lithium ion batteries causes less harm to the environment than that of nickel-cadmium batteries.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Practice paper 1A, 2012, 23)

15 Which of the following statements about a nickel metal hydride cell is / are correct?

(1) Its maximum voltage is 1.2 V.
(2) Its electrolyte is sulphuric acid.
(3) Its delivers relatively low discharge currents.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

16 Consider the following design criteria for primary cells. Each of them is crucial in certain uses. Suggest ONE use for each criterion.

a) The weight of a small portable power source is more important than its volume.

b) To give a steady voltage over continuous discharge.

c) No leakage is allowed.

17 Lithium ion cells are used in mobile phones.

a) List the materials used to make the anode and cathode of a lithium ion cell respectively.

b) What is the electrolyte in a lithium ion cell?

c) Suggest TWO advantages of using lithium ion cells in mobile phones.

d) Suggest ONE other device commonly operated using lithium ion cells.

18 Nowadays, lithium ion (Li-ion) batteries are more commonly used than NiCd batteries in portable electronic devices. Suggest ONE advantage of using Li-ion batteries over using NiCd batteries.

(HKALE, Paper 2, 2012, 7(c))

19 The table below includes some information about three types of AA-sized cells.

Type	Price per cell (HKD)	Shelf life (years)	Service life (minutes)
Zinc-carbon cell	2.0	1.5	50
Alkaline manganese cell	6.0	3	90
Nickel metal hydride cell	24.0	1	service life is not given here as this is unimportant for a rechargeable cell

Cost of a battery charger = HKD 115

The cost for charging an AA-sized nickel metal hydride cell is very low, usually about HKD 0.2–0.3.

(The service life of a cell has been determined from its use in a motorized toy.)

a) What is the maximum voltage of an alkaline manganese cell?

b) Decide and explain which type of cell you would use in a portable CD-player.